

ColorGuardian: Customize Skin Tattoos for Children with Vitiligo

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ABSTRACT

Vitiligo is a common pigmentary skin disorder. Children with vitiligo are likely to have low self-esteem and fear of social communication. The common treatment of vitiligo for children is physiological, rarely in psychological measures. In this paper, we present ColorGuardian, a new and interesting skin tattoo customization system, to relieve the psychological damage caused by vitiligo to children. The system scans the vitiligo area with the Light Detection and Ranging scanner to generate the 3D skin surface model and obtain the white patches image. Users can design their own tattoos or choose recommended designs that are available in the gallery to match up the pattern and the white patches' shape through the application that is linked to the system. We evaluate our system through a light usability test and semi-structured interview. The results show that ColorGuardian has positive impacts on children by reducing their feelings of inferiority.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools; User interface toolkits.

KEYWORDS

Skin tattoos, vitiligo children, tattoo printer, intelligent match, positive psychological intervention

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1 INTRODUCTION

Vitiligo, a depigmenting skin disorder, is characterized by the selective loss of melanocytes, which in turn leads to pigment dilution in the affected areas of the skin [1]. Vitiligo is prevalent in 0.04–2.2% of children [11]. Children with vitiligo (CWV) are prone to develop inferiority and social anxiety due to immature psychological development. Moreover, vitiligo's negative emotions are likely to cause the patients' immune system's disorder, further developing and spreading the white patches bring difficulties to the treatment [1]. Therefore, physiological treatment and positive psychological intervention [2] are indispensable to treat vitiligo in children.

Positive psychological intervention is not only helpful for the treatment of vitiligo disease but also can alleviate the psychological damage of CWV. At present, doctors have developed methods such as using hormone drugs to regulate the patient's immune system and laser therapy to guide pigmentation in clinical treatment [4]. However, the psychological treatment of CWV is often neglected. Psychological interventions for children mainly rely on professional psychologists to conduct professional psychological interviews or games. There are very few psychological interventions that are interesting, personalized, and involve the CWV actively. In this regard, we propose the ColorGuardian system, which uses customized tattoos to cover the vitiligo patches on CWV. Personalized tattoo patterns can be targeted to cover skin patches and attract children to participate in the activity. Hopefully, children can reduce their

sense of inferiority in social interaction through such an interesting camouflage.

Tattoos are a typical decoration on the skin. Compared to traditional tattoos, temporary tattoos are easy to remove and do not damage the skin. Temporary tattoos mainly include painted tattoos, sticker tattoos, and inkjet tattoos. Among the different types of temporary tattoos, inkjet tattoo has more advantages in terms of time cost and personalized design. Existing inkjet tattoo printers [12] can print various tattoos on the skin and be erased with alcohol or soda water. Inspired by that, the ColorGuardian system uses an inkjet tattoo printer to print custom tattoo patterns that support children's diverse designs without harming their skin.

In this paper, we propose a ColorGuardian system, a skin tattoo customizing and printing system for CWV. It can scan the vitiligo area to generate 3D skin models and obtain the white patches image of this part. To get a better match with patches' shapes, the pattern gallery in the software will recommend similar patterns, and users can also design tattoos themselves. The generated tattoo patterns can be printed on the skin by the ColorGuardian printer. Compared to previous printers, our printer is more interactive and can recognize vitiligo patches and print patterns on the corresponding position of skin. ColorGuardian is a digital method to make the camouflage of CWV interesting and accessible. It can achieve effective and positive psychological intervention for CWV. We conducted a usability test and a semi-structured interview to assess the system's effectiveness and user experience. Our main contributions are as follows: 1) We present a new skin tattoo customization and printing system, ColorGuardian for CWV, and develop the preliminary prototype; 2) We have verified the effectiveness of ColorGuardian, and the results show it has a positive psychological impact on CWV.

2 RELATED WORK

It has been proved that positive psychological intervention methods can reduce children's psychological challenges and benefit in their recovery. Maru E. Barrera proposed music therapy for preschool children and adolescents suffering from tumors and proved that music therapy effectively alleviated the anxiety in hospitalized children with cancer [3]. Marco Nalin uses companion robots to reduce stress in children with diabetes, and help them improve their response to treatments [15]. British Open Binoic's Hero Arm project [16] used "Star Wars", "Marvel", and "Frozen" as elements to customize princess-themed and hero-themed prostheses for children with physical disabilities, turning them into bionic heroes. Kids love the Hero Arm's cool appearance and feel less socially awkward when using it. It can be seen that introducing children's favorite elements into the research and design can contribute to achieving positive psychological intervention for children.

Multiple studies have shown that camouflage can improve patients' quality of life with vitiligo [[7], [17]]. Doctors recommend camouflage when patients with vitiligo are unwilling to be treated, or the treatment fails to work as expected. The Medical School of Kyoto University in Japan had set up courses to teach patients how to cover vitiligo effectively [21]. Miki Tanioka presented makeup tips for camouflaging vitiligo of the fingers, using the proprietary acrylate terpolymer compound Cavilon No Sting Barrier Film (3M,

St Paul, Minnesota) as a spray [22]. Although high-quality cosmetics and tools can help achieve coverage, the precise cosmetic process is complicated and tiresome for children. Besides, it is a challenge for CWV to choose a suitable cosmetic color that matches their skin's colors.

The vitiligo patches' surface of CWV is smooth without peeling, and the skin does not shrink. Except for the different colors, it is no different from the normal skin around it [1]. This is suitable for printing tattoos on the skin to cover up patches. In the field of human-computer interaction, the concept of colored tattoos has already been applied to children's products. For example, Bongkeum Jeong used colored tattoo bracelets to record children's vaccination information in developing countries [10]. Renata Souza Luque used colored tattoo stickers to guide children's insulin injections to reduce their resistance to injections [13]. Previous studies have illustrated that children express greater acceptance and affection for products incorporating colored tattoos. In this paper, we introduce the colored tattoo with child-friendly ink to cover the vitiligo patches of CWV, and we hope that the colored tattoo can help CWV accept the white patches on their bodies.

3 SYSTEM DESCRIPTION

The ColorGuardian system consists of two parts: a tattoo printer and a pattern customization application. The tattoo printer includes an inkjet system, camera, LCD, displacement sensor, LED light, and start button (Figure 1). The printer's printhead size is 14mm, which means the maximum width of each print pattern is 14mm, so we need to print step by step for the pattern whose width exceeds 14mm. With the handheld printing way, users can easily print patterns on various parts of their bodies. 3D printer and Raspberry Pi were used to develop the printer module.

To make a better the tattoo pattern to cover the vitiligo patch, we need to obtain the size information of the skin patch. Therefore, we use the Light Detection and Ranging (LiDAR) scanner installed on the iPad Pro 2020 to quickly generate 3D skin models and obtain the white patches image. The generated 3D models can be present in the ColorGuardian App based on the iPad. Users can select the pattern recommended by the gallery and adjust the pattern size to cover the white patches or draw personalized patterns on the App with Apple Pencil.

The using process of the ColorGuardian can be divided into three steps: 1) Body scanning and 3D skin models processing. 2) Tattoo pattern customization. 3) Skin tattoos printing.

3.1 Body Scanning and 3D Skin Models Processing

Firstly, users need to scan the body part with white patches by the 3D scanner LiDAR and 3D Scanner AppTM on the iPad (Figure 2a). The resulting data can be uploaded to the server. Then we select and cut the effective region to get the target 3D surface model on the server-side. Next, the QuadriFlow algorithm [5] will be used to mesh the 3D surface. According to the paper [14], we control the number of faces to about 25 after remeshing. After that, the remeshing 3D surface can be flattened into a 2D pattern using origami-based flattening algorithms [9]. Based on the 2D pattern, we use the OTSU algorithm [18] to identify the edge of the vitiligo

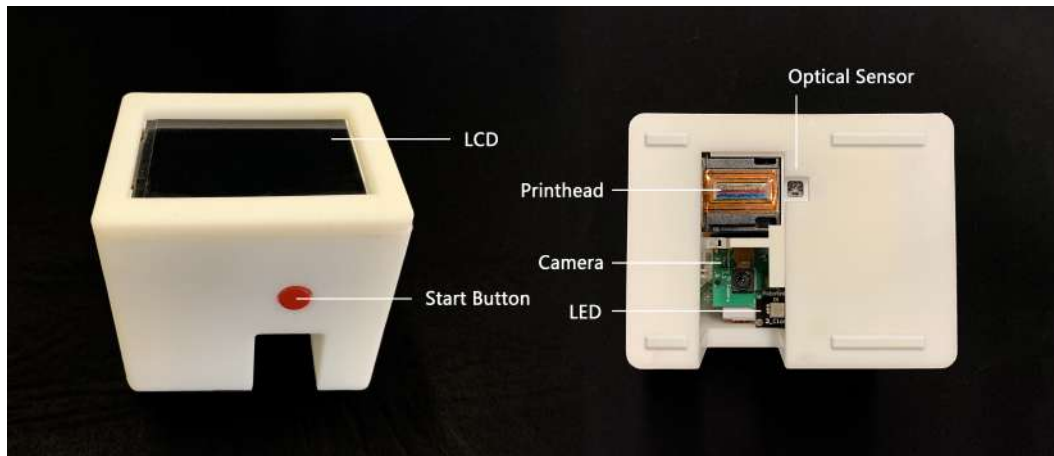


Figure 1: The tattoo printer of the ColorGuardian system



Figure 2: Usage steps of ColorGuardian:(a) Body Scanning;(b) Tattoo Pattern Customization;(c) Skin Tattoos Printing;(d) Finish.

patch on the map to obtain the white patch's size information and return the final output image to the ColorGuardian Application.

3.2 Tattoo Pattern Customization

In the ColorGuardian App, after selecting the 2D pattern of patches, users can draw personalized patterns to hide patches (Figure 3b) or choose patterns from the gallery recommendation (Figure 3a). The images in the gallery come from the Cartoon Dataset we collected from the open-source graphics website [20]. We categorize all the cartoon images into themes of "Superhero", "Princess", "Animal", "Plant", "Ocean", "Monster" and "Other". Using Hu invariants [6], the application matches all patterns that are most similar to white patches from the dataset and recommends them to users.

In the print preview interface (Figure 3c), users can preview the effect of the tattoo pattern printed on their skin. If they are not satisfied with the customization, they can go back to the previous step and continue with the design. After being satisfied, users upload the tattoo pattern. ColorGuardian Application will cut the pattern into several parts based on the print width of 14mm (Figure 3c). Then users can print the pattern according to the order prompted by the application.

3.3 Skin Tattoos Printing

The tattoo pattern data is transferred to the printer via Bluetooth. The tattoo printer uses child-friendly ink, Prinker Ink [19], which is FDA VCPR and EU CPNP compliant. To reduce printing errors, several measures have been taken. The camera and LCD on the printer can help users confirm the print location. The camera's height is 50mm, and the distance between the printhead and the camera is 25mm. When the printer is placed on the skin, the camera can capture the range of 35*35mm image below, and LED can supplement light for the camera. We also provide print guidance on the LCD. As shown in Figure 4, the blue area on the LCD represents the printhead's location and its printing area. The red line is the white patch's outline identified from the model map. Users need to adjust the printer's position until the white patch area on the LCD captured by the camera can match the red line (Figure 4). After determining the correct printing position under the guidance, users can press the printer's start button and move the printer rightward to print the tattoo (Figure 2c). The optical sensor can also detect the printed length to make sure the tattoo is printed in the corresponding position.

4 USER STUDY

To explore the effectiveness and user experience of ColorGuardian, we conducted a user study within the usability test and semi-structured interview. This study recruited six children aged 5-10 years with mild vitiligo (two males and four females, mean age=6.83, SD =2.14). Their white patches were all smaller than 12cm² and distributed on the arm, neck, and calve. These six children have

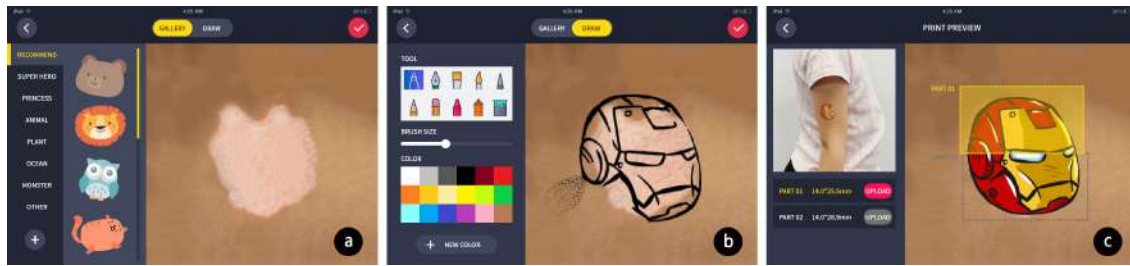


Figure 3: Interfaces of ColorGuardian App: (a) Choose patterns from our gallery; (b) Draw custom patterns; (c) Preview the pattern on the skin.

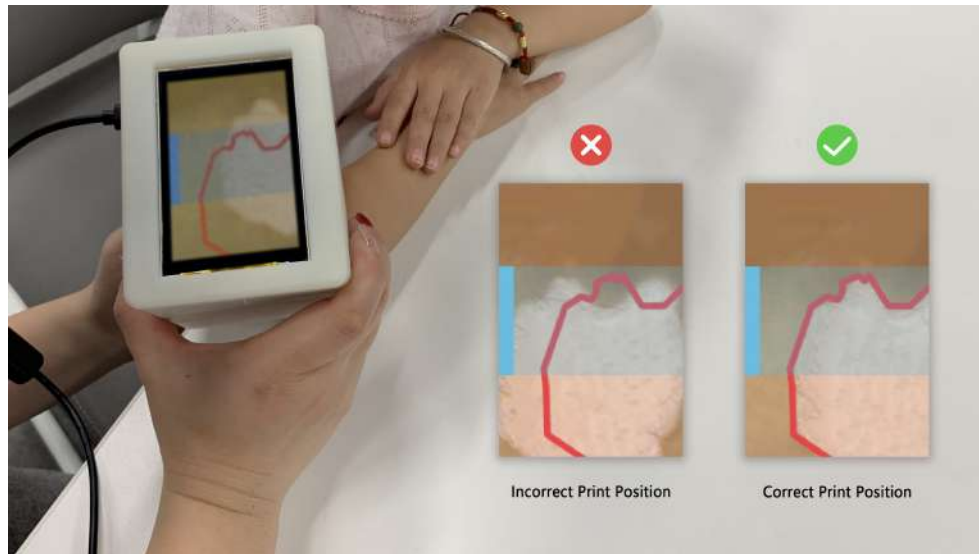


Figure 4: print guidance on the LCD of the printer

taken medication to prevent white patches' spread but had no laser treatment or cosmetics experience to cover patches. All children's parents have approved the experiment and signed informed consent forms. We also consulted a dermatologist with 10 years of experience. With his guidance and assist, six participants completed the skin allergy test to printer ink. Tests showed that none of the six participants were allergic to the ink used. The whole experiment was recorded with the consent of the child's parents.

4.1 Procedure

Firstly, all participants completed a questionnaire survey about the impact of vitiligo on children's social communication. The questionnaire was modified based on the Children's Dermatology Life Quality Index (CDLQI) [8]. Within 10 minutes, the children were asked to complete the questionnaire on their own, and for those who could not read independently, researchers would help them complete the questionnaire.

Then we introduced participants to the ColorGuardian system for 10 minutes. In the next 30 minutes, the participants used the ColorGuardian system to customize and print a tattoo on their skin patches. We recorded children's using process and interactions with

the ColorGuardian. Also, we observed their facial expressions and conversations with their parents. After finishing the tattoo printing, we asked participants to wear the color tattoos for 1 day of free activities. In the next afternoon, we conducted semi-structured interviews with the participants and their parents. The semi-structured interviews focus on three questions: 1) For Children: Are you interested in the ColorGuardian system? 2) For Children: Does the ColorGuardian make you feel confident in social life? 3) For Parents: How do you think the ColorGuardian and do you agree with your child using this system? We also conducted an interview with the dermatologist to understand his views on ColorGuardian.

4.2 Result

We found that all participants successfully completed the personalized skin tattoo printing with the help of the researcher and did not show any physical or psychological discomfort. We analyzed the data from questionnaires and user interviews, the results came out as following:

4.2.1 Questionnaire. The questionnaires showed that all participants had the experience of being asked questions by other children about their vitiligo patches, which made them embarrassed. One

child said he had been nicknamed because of vitiligo, and it had affected his friendships. Another child said the patches on his skin were ugly, and he didn't like them. Four children said they would wear special clothes to hide the white patches. As a result, vitiligo has affected the children's social life and sometimes leaves them feeling inferior and embarrassed.

4.2.2 Semi-structured interviews. All the participants said ColorGuardian was an interesting system, and they showed a high willingness to use the system. During the experiment, one girl chose to draw the tattoo pattern, while other children selected their favorite tattoo patterns from the gallery recommendation. After finishing the pattern printing, the participants were excited and would intuitively show the skin tattoos to their parents. Furthermore, they were delighted to print tattoos and would like to use ColorGuardian again to cover the white patches in the future.

Participants also present confidence in social interaction after using the ColorGuardian system. One boy mentioned his experience when he showed his friend the "Pikachu" (an animated role) pattern on his arm: "my friend said it was cool, and he also wanted one." Another child said he did not worry about new friends asking about the white patches' questions. With the help of ColorGuardian, children felt more socially confident, and they would like to introduce skin tattoos to others.

We also tested parents' views of ColorGuardian. All parents agreed with their children using the system and recognized the positive psychological effects of ColorGuardian on their children. For example, one mother said that her child liked the tattoo and was actively showing it to family and friends and taking photos to commemorate it. Another mother said she looked forward to seeing ColorGuardian bring more positive experiences to children.

In our interview with the dermatologist, we learned that he thought the ColorGuardian system could help CWV to improve their confidence and that occasional use of the child-friendly ink would not affect their health. However, as a long-term cover for CWV, it needs to be tested in further experiments.

Overall, we found that children expressed a strong interest in ColorGuardian to print tattoos covering white patches. Both children and parents thought the ColorGuardian system is effective and can reduce the inferiority feeling of CWV and bring them positive psychological influence.

5 LIMITATIONS AND FUTURE WORK

During development and experimentation, we realized the limitations of our prototype. At present, our system relies on the iPad's LiDAR scanner and 3D Scanner AppTM to scan body parts. The software has limited scanning accuracy that will affect the 3D model. Besides, we used the Raspberry PI component to develop the hardware, making the printer's size large enough for children to print tattoos themselves. So that the parents or researchers need to assist the printing process. When printing the tattoo pattern, whose width exceeds 14mm, we use a step-by-step printing method. Printing errors probably occur due to users' operation even though we have used the camera and LCD to guide printing visually.

In the future, we will optimize the accuracy of printing and improve the prototype. On top of that, we will enrich the patterns of the gallery to meet children's diverse needs and more accurate

pattern matching. Furthermore, an extensive range of qualitative and quantitative user studies are required on CWV to improve our design rigorously.

6 CONCLUSIONS

In this paper, we developed ColorGuardian, an interesting tattoo pattern customization and printing system for CWV. It supports children's personalized tattoo design and can effectively hide white patches on the skin. We conducted studies to test the system's effectiveness and user experience. The results show that CWV are interested in the ColorGuardian system and are willing to use it. They have reduced the inferiority in social interaction and want to share the skin tattoo with others. The ColorGuardian has also been recognized and expected by parents preliminarily. Overall, we hope ColorGuardian can help CWV and give them positive psychological intervention. Besides, it is expected that the system will inspire researchers, parents, and our society to focus on the psychotherapy of CWV.

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